

Iker Garcia Morales

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Electrical Engineering student with hands-on experience in digital design, VLSI layout, and hardware verification.

Skilled in Verilog/VHDL, CMOS layout (Electric VLSI), and simulation tools (ModelSim, Quartus).

Interested in CPU architecture, ASIC design, and next-generation semiconductor systems.

SKILLS:

Hardware Description Languages: Verilog, VHDL

Programming: C, C++, Python, MATLAB

VLSI & Digital Design: CMOS Layout (Electric), Digital Logic Design, ASIC Design Flow

Simulation & Tools: ModelSim, Quartus Prime, LTSpice

Embedded Systems: Arduino, ESP32

Other: Linux, Git, Simulink

EDUCATION:

John Brown University, Siloam Springs, AR

Bachelor of Science in Electrical Engineering, Double Major in Robotics and Mechatronics Engineering.

Expected Graduation: May 2027.

Relevant Coursework

Integrated Circuit Design: Electric VLSI, Embedded Systems, Electromagnetics, Electronics, Electrical Circuits, Digital Electronics, Control Systems.

PROJECTS:

VLSI Design Project: 8-bit MIPS Processor (Electric VLSI)

Designed and implemented an 8-bit MIPS CPU using a bitslice architecture, including datapath, control unit, and ALU.

Performed full-custom CMOS layout, passing DRC/ERC/NCC verification.

Simulated functionality using IRSIM and validated correct operation across multiple test scenarios.

Integrated core into padframe for chip-level design targeting MOSIS TinyChip process.

Implemented control logic using FSM-based controller and integrated datapath modules across 8-bit slices.

Embedded Systems: Ground Recognition Rover (ESP32-Based):

Developed dual-board system using ESP32 and ESP32-CAM, Integrated BME280 environmental sensor and L298N motor driver, Implemented wireless control and telemetry via Wi-Fi.

Digital Electronics:

Designed and simulated a serial 4-bit ALU at gate level using Quartus, implementing arithmetic and logic operations (add, subtract, XOR, increment). Developed sequential architecture using shift registers and carry logic, and verified functionality through simulation and timing analysis.

Electronics (Buck Converter PCB Design)

Designed and built DC-DC step-down converter, calculated duty cycle and component values, designed custom PCB layout, validated output voltage regulation and efficiency under load

RELEVANT WORK EXPERIENCE:

John Brown University — Siloam Springs, AR

Teaching Assistant – Digital Electronics

Jan 2025 – Present

Facilitated lab sessions on digital circuit design and troubleshooting, supported students in logic gates, flip-flops, multiplexers, and counters, graded assignments emphasizing circuit correctness and technical precision

Learning Coach – Student Support Services

Aug 2025 – December 2025

Provided structured academic support through one-on-one and small group sessions, strengthened student problem-solving skills and study strategies in technical coursework

General Tutor (Engineering & Math)

Sept 2024 – Present

Tutored Calculus and Electrical Circuits, reinforced circuit analysis techniques and mathematical foundations for engineering students

LEADERSHIP EXPERIENCE:

IEEE Club – Junior Senator

Student Government Association – Freshman Senator, Junior Senator.

Walton International Scholarship Scholar.